

Ahmed Azaz Humdoon

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RESEARCH INTERESTS

- AI for Software Engineering
- Trustworthy & Robust AI
- Reliable LLM-based Systems
- Software Testing with LLMs

EDUCATION

University of Central Florida <i>Ph.D. in Computer Science and Engineering</i> • CGPA : 4.00/4.00 (First semester)	Orlando, Florida, USA. <i>Jan. 2026 – Present</i>
Islamic University of Technology <i>Bachelor of Science in Computer Science and Engineering</i> • CGPA : 3.58/4.00	Dhaka, Bangladesh. <i>Jan. 2018 – May 2022</i>
Notre Dame College <i>Higher Secondary Certificate</i> • GPA : 5.00	Dhaka, Bangladesh. <i>2015 – 2017</i>

PUBLICATIONS

Ahmed Azaz Humdoon, Cheng Chu, Lei Jiang, Qian Lou, Mengxin Zheng. “SoK: Adversarial Robustness of the Variational Quantum Eigensolver via Red-Teaming.” *Q-Sec Workshop, IEEE QCE 2026*. (Accepted)

RESEARCH EXPERIENCE (Undergraduate Thesis) [Preliminary Draft](#) | [Thesis Book](#)

Semi-Automated Approach to Generate Bangla Dataset for QA & Query-Based Text Summarization

- We provide a novel unified pipeline to generate dataset for three tasks : Question-Answering (QA) and Query-based Single & Multi Document Summarization (SD-QBSUM, MD-QBSUM).
- Incorporated a semi-supervised approach to annotate data in a semi-automatic manner. Our work primarily involves the Google NQ dataset.
- We incorporate HITL (Human-in-the-loop) to ensure the high quality of the dataset.
- This work was funded by Institute for Advanced Research, United International University (Co-Investigator, BDT 489,000; Grant Ref: UIU-IAR-02-2022-SE-07).

INDUSTRY EXPERIENCE

Software Development Engineer July 2023 – Dec 2025

IQVIA Ltd

- Delivered impactful user features on time for multiple applications, prioritizing best engineering practices for optimal performance and maintainability.
- Designed and developed responsive client-facing interfaces using TypeScript, Angular, CSS, SCSS to facilitate user interaction.
- Enhanced the stability and reliability of multiple apps by significantly expanding test coverage.
- Actively participated in technical discussions with project managers, team leads, quality assurance engineers, and other developers to collaborate on devising viable solutions.
- Contributed to building reusable code and libraries for future use.
- Performed peer code reviews to ensure clean coding practices and maintainable design.

Assistant Software Engineer January 2023 – June 2023

Millennium Information Solution Ltd

- Developed RESTful APIs to manage authentication, account information, billing and financial transactions of DESCO (Dhaka Electric Supply Company).

- Contributed to developing the official documentation website of the company using Docusaurus which gives full details about the technological architecture of how the softwares are made.
- Gained knowledge about microservice architecture and CQRS design pattern.

Software Engineer Intern

March. 2021 – October. 2021

Business Automation Ltd

- Worked on developing one in-house information management & analysis system application using Flutter.
- Learned Clean Code, practiced Object Oriented Design SOLID principles and implemented different Design Patterns such as Factory, Singleton, State, Prototype pattern in Java.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++.

Deep Learning Frameworks & Libraries: PyTorch, Tensorflow, Keras, Scikit-Learn, NumPy, Pandas, NLTK.

Web Development: HTML, CSS, JavaScript, TypeScript, Angular.

Database Management & Other Tools: MySQL, PostgreSQL, Git, LaTeX, Hugging Face, Jira.

SELECTED PROJECTS

Hallucination Detection and Measurement in LLMs | *Python, OpenAI/Google/Groq APIs* |

- Built a systematic benchmark measuring hallucination across seven LLMs (three closed-source, four open-source), three task types (factual QA, logical reasoning, summarization), and two prompt designs, collecting and analyzing 3,500 responses.
- Detected hallucinations using reference-based ROUGE-L scoring and answer-correctness verification through task-specific evaluation pipelines.
- Found task type to be the strongest factor (factual QA highest at 42.7%, summarization lowest at 13.6%), with few-shot prompting failing to reliably reduce hallucination over zero-shot.

Spam Message Classification using Naive Bayes Algorithm | *Python, Scikit-learn, Nltk, Pandas* |

- Spam message classification is performed using the Naive Bayes algorithm, leveraging Scikit-learn, NLTK, and Pandas for implementation.

Text Summarization with BART and T5 | *Python, Tensorflow* |

- Deep Learning project for text summarization using BART and T5 from Hugging Face Transformers library.

Offensive language identification from tweets using BERT | *Python, Tensorflow* |

- Natural Language Processing (NLP) application on determining whether a social media tweet/post is offensive or not.

Chess Game | *C++, graphics.h library* |

- A classic two player chess game that ends when a player is first checkmated.
- graphics.h library was used for creating the interface.
- The game is continued till one of the player doesn't play more than 50 moves.

CERTIFICATIONS

PyTorch Ultimate 2024: From Basics to Cutting-edge | *from Udemy [Verify]*

Deep Learning Specialization | *from Deeplearning.ai [Verify]*

EXTRACURRICULAR ACTIVITIES

IUT Mars Rover Team, Autonomous and Communication System

European Rover Challenge - 2020

- Our team achieved 19th position in the competition and 2nd position among all Bangladeshi teams.

IUT Computer Society, Event Management Team

10th IUT ICT FEST - 2019

- Involved in management of event scheduling and team registration for different programs of the fest.
- Volunteer for the Application Development hackathon event .